

From the Cover: Transfer of a cyanobacterial neurotoxin within a temperate aquatic ecosystem suggests pathways for human exposure

β -methylamino-L-alanine (BMAA), a neurotoxic nonprotein amino acid produced by most cyanobacteria, has been proposed to be the causative agent of devastating neurodegenerative diseases on the island of Guam in the Pacific Ocean. Because cyanobacteria are widespread globally, we hypothesized that BMAA might occur and bioaccumulate in other ecosystems. Here we demonstrate, based on a recently developed extraction and HPLC-MS/MS method and long-term monitoring of BMAA in cyanobacterial populations of a temperate aquatic ecosystem (Baltic Sea, 2007–2008), that BMAA is biosynthesized by cyanobacterial genera dominating the massive surface blooms of this water body. BMAA also was found at higher concentrations in organisms of higher trophic levels that directly or indirectly feed on cyanobacteria, such as zooplankton and various vertebrates (fish) and invertebrates (mussels, oysters). Pelagic and benthic fish species used for human consumption were included. The highest BMAA levels were detected in the muscle and brain of bottom-dwelling fishes. The discovery of regular biosynthesis of the neurotoxin BMAA in a large temperate aquatic ecosystem combined with its possible transfer and bioaccumulation within major food webs, some ending in human consumption, is alarming and requires attention.